



— PIGGING SOLUTIONS —

PIGGING VALVE

USAGE, INSTALLATION & APPLICATION GUIDE

for Pipeline Pigging Systems

2-WAY | 3-WAY | 4-WAY

Model: LNS-WPV



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Prepared By : LINESIG Engineering
Approved By : _____



RELIABLE



PRECISE



DURABLE



SAFE

RELIABLE PIGGING. SMOOTH OPERATION. SAFE PERFORMANCE.

DATA SHEET | LNS-PW DN100 TO DN1200

Positive Pig Routing. Reduced Station Complexity.

A heavy-duty three-port pipeline valve engineered to route process flow and pigs between a common line and either of two branches. Designed for terminals, pipeline stations, oil-feed lines, tank farms, refineries, offshore systems and utility networks.

- DN100-DN1200 nominal range
- Piggable selected flow path
- Three-port diversion with positive branch selection
- Hydrocarbon, gas, water, chemical and marine metallurgy options
- Manual, electric, pneumatic or hydraulic actuation



1. Application

LINESIG LNS-PW 3-Way Pigging Valves are intended for controlled diversion, pig launching/receiving routing and transfer service in pipelines carrying crude oil, refined petroleum products, natural gas, fuel gas, water, chemicals and other compatible fluids. Final valve configuration is selected according to the working medium and project operating conditions.

2. Connection, Installation & Environment

Item	Standard / Available Arrangement
Installation Position	Normally aboveground; preferred actuator/gear orientation upward or as approved by GA
Pipeline Connection	Butt-weld ends, flanged ends or project-specific connections
Pup Pieces	Can be supplied separately or integrated/welded where specified
Ambient / Climatic Duty	Temperature, cold or warm-climate construction is available by material, seal and actuator selection
Process Temperature	Project-specific; standard material/seal package selected for required range
Support	Integral support legs / structural support provisions subject to size and GA

3. Tightness & Isolation Philosophy

The selected branch is intended to provide positive isolation of the non-selected path while maintaining the designed piggable flow path. Seat design may use pressure-assisted and spring-loaded sealing principles with soft or engineered seats selected to suit service.

ZERO-LEAKAGE OPTION:

High-tightness / zero-visible-leakage shutoff can be specified where the selected seat design, pressure class and project valve standard support requirement.

4. Design Features

Design Feature	LINESIG Configuration / Benefit
Trunnion-Supported Rotary Element	Preferred for large diameter/high torque service; reduces stem loading and stabilizes internal rotating element.
Split-Body Construction	Improves maintainability and enables access to internal components without replacing the complete valve assembly.
Bypass / Equalization Cavity	Optional engineered bypass path can maintain limited process circulation during selected operating stages.
Drain & Vent	Body/cavity drain and vent connections for depressurization, condensate removal and maintenance.
Pressure Relief Provision	Relief connection/valve option for safe cavities or chamber pressure management.
Sealant / Grease Injection	Optional emergency or maintenance injection points for stem/seat sealing systems where applicable.
Piggable Bore	Selected port is matched to pipeline ID and pig geometry to avoid snagging or excessive restriction.
Position Indication	Clear local branch-position indication with optional electrical feedback.

SPLIT BODY PIGGING VALVE:

Split-body design increases valve maintainability, allows to use it at aggressive working medium and conduct: components replacement; maintenance without dismantling.



BYPASS CAVITY:

Pigging valve ball has bypass cavity. It ensures working medium flow through ball valve in "closed" position (At the moment of loading or removing a pig). Bypass cross section is about 25% of pipeline cross-section. Herewith medium pressure in the pipeline is not changed.

NOTE: Ball valve can be supplied without bypass cavity upon customer's request



TRUNNION MOUNTED BALL:

Pigging valve are developed on basis of PTPA high-performance trunnion mounted ball valve having the same design, features, performance, reliability, operability and material selection



SAFETY:

Launching and receiving cameras are equipped with pressure relief valves to ensure safe operation of the Pig Valve.



DRAIN & VENT:

Drain connections are installed at ball valves for removal of condensate fluid from the body.

5. Construction & Material Options

Code	Component	Typical Material Options	Selection Basis
1	Body / Main Pressure Parts	ASTM A216 WCB; A352 LCB/LCC; A351 CF8M; duplex/super duplex	Pressure, temperature, corrosion, low-temperature or marine duty
2	Rotary Ball / Diverter Element	Carbon/low-temp steel with overlay; SS316; duplex	Pig contact, erosion, corrosion and pressure class
3	Stem / Trunnion	A182 F316; 17-4PH; duplex / engineered alloy	Strength, corrosion and torque
4	Seats / Seat Holders	Carbon/low-temp steel, SS316 or CRA; replaceable insert where selected	Shutoff class, service and maintainability
5	Seat Insert / Primary Seal	RPTFE, PEEK, polyurethane/elastomer or engineered soft seat; metal seat optional	Pressure, temperature, pig duty and medium
6	Bolting	A193/A194; A320/A194 for low-temperature; stainless/duplex equivalents	Temperature, pressure and environment



6. Material Selection by Service

Service	Body / Pressure Parts	Trim / Wetted Parts	Seat / Seal Direction
Crude Oil / Oil Feed	A216 WCB	SS316 / 17-4PH or engineered trim	FKM, HNBR, RPTFE / PEEK
Refined Products / Diesel / Jet Fuel	WCB or CF8M	SS316	FKM, RPTFE / PEEK
Natural Gas / Fuel Gas	WCB; LCB/LCC for low-temperature duty	SS316 / 17-4PH	HNBR/FKM or low-emission package
Sour Hydrocarbon	WCB/LCC or CRA configuration to approved sour-service metallurgy	SS316/duplex/nickel-alloy as required	H2S/hydrocarbon-compatible sealing
Produced Water / Industrial Water	Coated WCB or CF8M	SS316	EPDM/FKM/PTFE as compatible
Chemicals / Petrochemicals	CF8M/316 or project-selected alloy	SS316/316L or higher alloy	PTFE/RPTFE/PEEK or compatible elastomer
Low-Temperature Hydrocarbon	A352 LCB/LCC	SS316	Low-temperature compatible seal system
Seawater / Offshore Utility	Duplex 2205 / super duplex by project metallurgy	Duplex / super duplex	Compatible PTFE/elastomer system

MATERIAL SELECTION:

Final materials must be confirmed against full fluid composition, chlorides, H₂S/CO₂, water cut, solids, design temperature, pressure, corrosion allowance and client metallurgy specifications.

7. Valve Advantages

- Reduces installation footprint compared with selected traditional launcher/receiver or crossover manifold arrangements.
- Reduces the number of separate isolation/diversion valves, tees, flanges and field welds in suitable configurations.
- Provides a dedicated pig-routing path and clear branch-selection philosophy.
- Large-diameter design can be integrated with gearbox or powered actuator for manageable operating torque.
- Split-body / serviceable construction can reduce maintenance scope where incorporated in the final design.
- Compatible with spherical, cup/disc, brush, cleaning and intelligent pigs subject to final bore and pig clearance study.

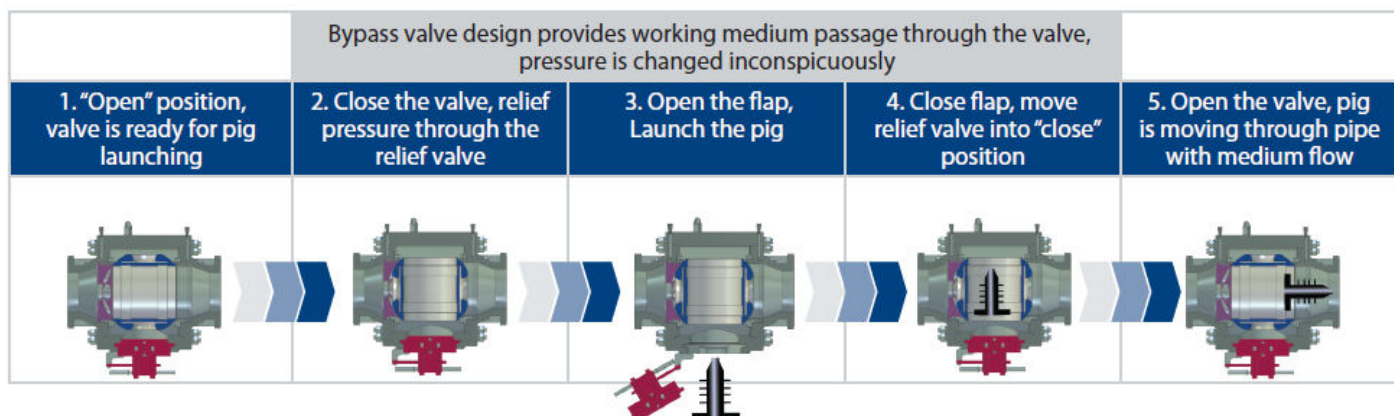
8. Conventional Manifold vs 3-Way Pigging Valve (Comparison)

Traditional Routing Arrangement	LINESIG 3-Way Pigging Valve Arrangement
Multiple isolation/diversion valves required	Single engineered three-port routing assembly
More tees, flanges, pup pieces and field welds	Reduced piping interfaces in suitable layouts
Larger station footprint	Compact valve/manifold footprint
More individual operating steps	Clear Position A / Position B branch selection
More valve items to maintain	Reduced valve count; service focused on one routing unit
Separate feedback from several valves	Centralized branch-position feedback

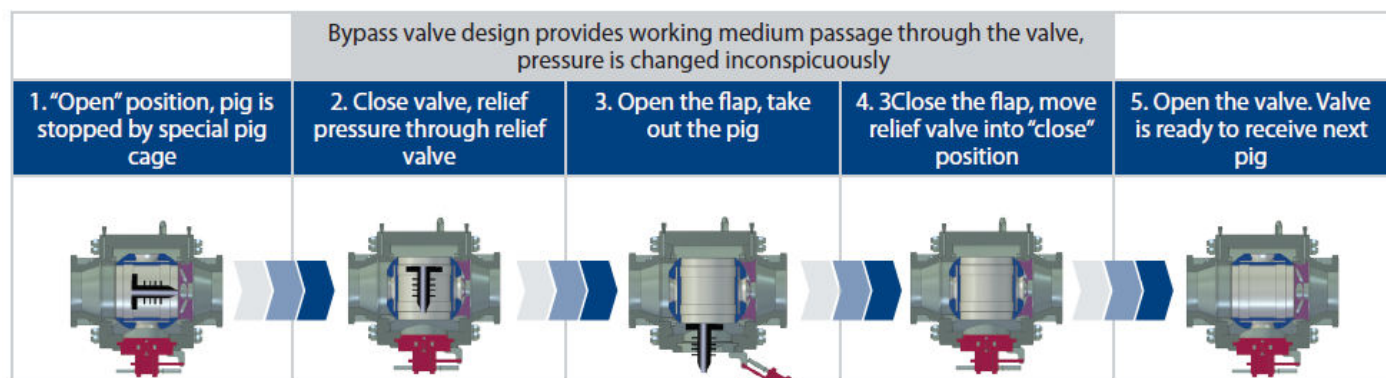
9. Operating Sequence – Typical

Stage	Typical Operating Action
Confirm Line-up	Verify selected pig route, valve position feedback and downstream readiness.
Equalize / Stabilize	Use approved bypass/equalization path where fitted and required by operating procedure.
Select Branch	Operate valve to connect common port with the required branch.
Confirm Isolation	Verify non-selected branch isolation and position indication/interlocks.
Launch / Receive Pig	Conduct pigging operation using the approved launcher/receiver procedure.
Restore Normal Routing	Return valve to required operating position and confirm remote/local status.

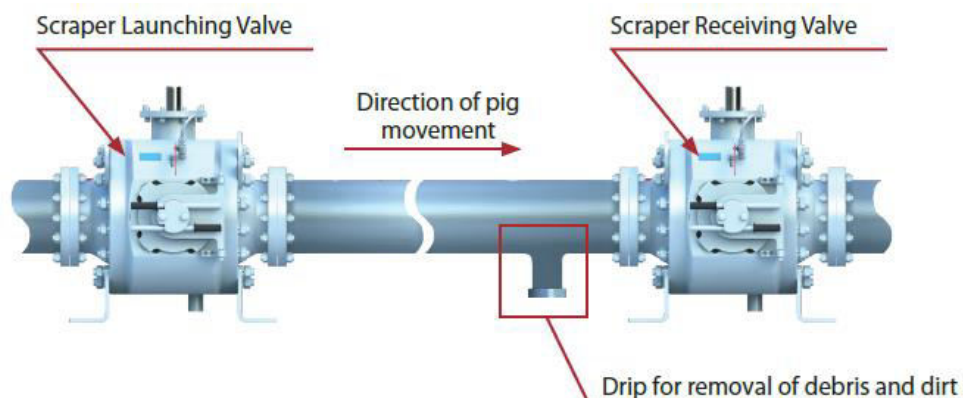
LAUNCHING:



RECEIVING:



10. Installation



11. Technical Range

Parameter	Engineering Range
Nominal Size	DN100-DN1200
Nominal Pressure	ASME pressure classes selected by project; high-pressure availability depends on DN and final design
End Connections	RF / RTJ flanged; butt-weld; special ends by project
Flow Function	Common port to Branch A or Branch B
Flow Direction	Bi-directional in selected piggable path, subject to approved design
Bore	Full / piggable bore to project ID and pig clearance requirements
Actuation	Gearbox, electric, pneumatic or hydraulic
Position Feedback	Mechanical indicator; limit/proximity switches; transmitter optional
Drain / Vent	Available as required
Bypass / Equalization	Optional engineered feature
Fire-Safe Configuration	Available where a tested/qualified design is specified
Low-Emission Sealing	Available where specified

12. Actuation & Control

Actuation	Typical Use	Control / Feedback
Manual Gearbox	Local operation / infrequent switching	Mechanical indicator, locking provision
Electric Actuator	Terminal/process automation	Torque/limit protection, local controls, PLC/DCS/SCADA feedback
Pneumatic	Automated plant service	Solenoid, switches, position transmitter, local panel
Hydraulic	High-torque / remote pipeline station	HPU, accumulators, ESD interface, local/remote control

13. Safety, Drain, Vent & Maintenance Provisions

Pressure relief connection/valve where cavity or chamber pressure relief is required.

Drain connection for condensate, debris or trapped liquid removal.

Vent connection for depressurization and safe maintenance.

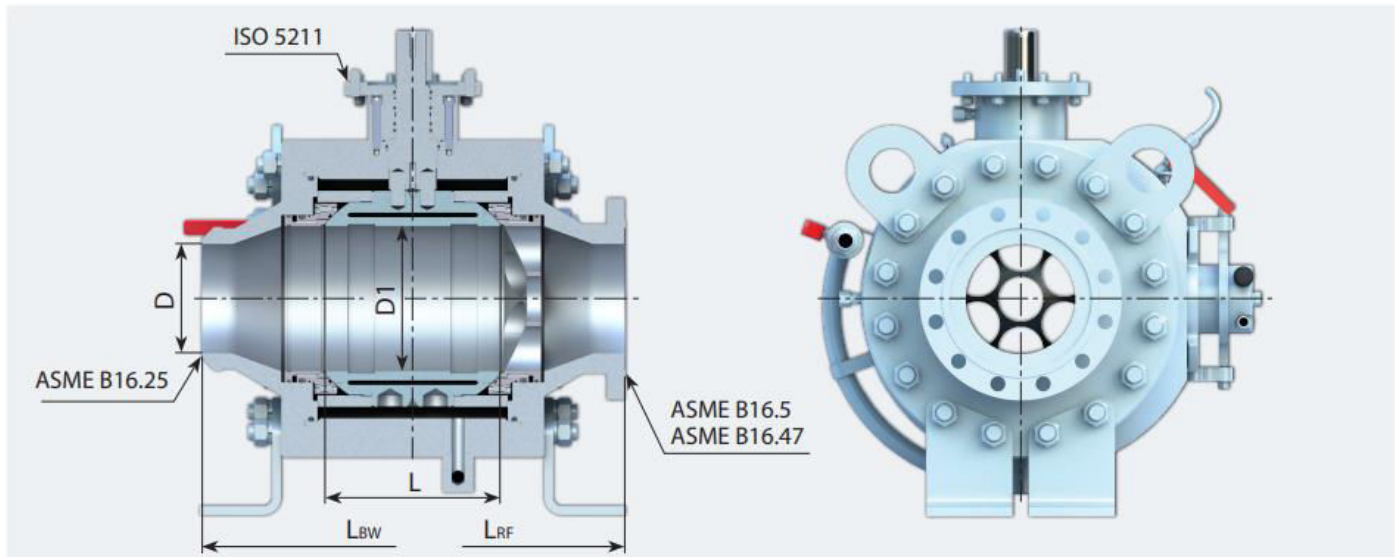
Optional sealant/grease injection to stem/seat systems where compatible with the selected design.

Mechanical or key interlocks to prevent unsafe branch selection or operation during pigging.

Lockable actuator/gear positions and local branch identification.

14. Main Dimensions – LINESIG Schedule

The supplied comparison catalogue gives a strong presentation format for dimensional data, but its actual dimensions apply only to its own NPS 6–16 product range and shall not be transferred to the LINESIG DN100–DN1200 valve. The following schedule is therefore provided as the editable LINESIG dimensional framework.



Reference dimensional-format illustration only — not LINESIG dimensions.

NPS	Series	Class	Dimensions, mm					Connection type according to ISO 5211	Weight*, kg
			D	D ₁	L	L _{BW}	L _{RF}		
6"	PT39140	150-600	150	185	261	813	660	F16	470
		800-900				889	737	F25	510
8"	PT39140	150-300	200	235	408	1016	794	F16	1170
		400						F25	
		600				1092	889	F30	
		800-900							
10"	PT39140	150-300	240	320	386	1194	940	F25	1635
		400-600						F30	1735
		800-900				1295	1041	F35	1850
12"	PT39140	150-300	300	340	499	1372	1067	F25	1885
		400-600						F30	
		900				1499	1194	F35	
14"	PT39140	150-300	830	487	160	1194	1194	F35	1869
		400	866	487	180	1321	1321	F40	2615
		600-900	968	589	98	1143	1067	F30	2420
16"	PT39140	150-300	400	438	892	1778	1372	F25	2720
		400						F30	
		600-900				1905	1499	F35	

15. Climatic & Environmental Configuration

Category	Available Engineering Direction
Temperate Climate	Standard outdoor industrial construction
Cold Climate	Low-temperature body material, seals, bolting, lubricant and actuator package
Warm / Desert Climate	High-ambient actuator/electrical package, UV/weather protection and project coating
Marine / Offshore	Enhanced external corrosion protection; stainless/duplex metallurgy as required
Hazardous Area	Certified actuator, switches, solenoids and accessories to project area classification

16. Inspection, Testing & Documentation

Category	Typical / Available Requirement
Material Traceability	EN 10204 3.1 certificates; PMI where specified
Pressure Test	Shell and seat testing to approved valve/project standard
NDE	VT, PT, MT, UT or RT as defined by material class and ITP
Dimensional Inspection	Bore/ports, connection dimensions, centerlines, support and actuator interface
Functional Test	Full travel, branch selection, feedback and interlock verification
FAT	Valve/actuator/control function and signal verification
Documentation	GA, datasheet, calculations as applicable, MTCs, NDE/test reports, ITP/QCP, coating report and O&M manual

17. Scope of Supply

- Fully assembled 3-way pigging valve in accordance with approved datasheet and GA.
- Selected gearbox or powered actuator with manufacturer documentation.
- Limit switches / position transmitter / solenoids / local control accessories as ordered.
- Support legs or mounting arrangement where included in approved scope.
- Commissioning spares / quick-wear parts kit as specified.
- Counter flanges, fasteners, gaskets or pup pieces when specifically included.
- Documentation and certification package.

18. Ordering / Selection Data

Required Customer Data	Information to Provide
Pipeline	DN/NPS, OD/ID, wall thickness, common and branch sizes
Process	Fluid, composition, density, viscosity, solids/water content
Pressure	Design/operating pressure and required class/rating
Temperature	Design minimum/maximum and normal range
Pig	Type, body/cup/disc diameter, length, weight and passage criteria
Materials	Body/trim/seat/seal requirements, corrosion allowance, sour-service requirement
Connections	RF / RTJ / butt-weld and dimensional standard
Actuation	Manual/electric/pneumatic/hydraulic, fail philosophy and operating time
Controls	Local/remote, switches, transmitter, solenoid, ESD, PLC/DCS/SCADA
Environment	Ambient temperature, offshore/onshore, area classification, coating category
Documentation	ITP, FAT, NDE, MTC, coating, GA, calculations and O&M requirements

19. Statement

LINE*Si*G LNS-PW, 3-Way Pigging Valves, DN100-DN1200, are engineered three-port pipeline valves for positive routing of process flow and pigs between a common line and two branch connections. The product can be configured for crude oil, refined products, natural gas, water, chemicals, sour service, low-temperature hydrocarbon and offshore environments, with pressure class, metallurgy, seats, seals and actuation selected to match the actual pipeline duty.

